Ela Aquino

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EDUCATION

University of Southern California - Viterbi School of Engineering Expected: May 2028
B.S. Computer Science | Incoming Spring 2027 Transfer

De Anza College September 2024 - December 2026
Computer Science for Transfer *GPA: 3.71/4.00*

SKILLS

Programming Languages: C++, Java, Python, JavaScript, TypeScript, HTML, CSS

Frameworks & Libraries: React, React Native, Node.js, Expo, PySide6, OpenCV, MediaPipe

Databases & Tools: PostgreSQL, Supabase, Git, GitHub, Vercel, Unity, Figma

Concepts: Authentication, Database Design, Object-Oriented Programming, Full-Stack Development, prompt engineering

PROJECTS

Scavenger Hunt Web App | React, TypeScript, Supabase, Vercel | Live Demo | GitHub May 2025 - Present
- Independently architected and developed a full-stack web application for LEGOLAND Discovery Center Bay Area, securing management approval for deployment to digitize guest scavenger hunt activities for **200+ weekly guests**.
- Implemented secure administrator authentication and database-level authorization using Supabase Auth and PostgreSQL Row Level Security (RLS), restricting privileged CRUD operations to authorized accounts.
- Developed a PostgreSQL-backed content management system using Supabase JavaScript client queries, enabling administrators to create, edit, publish, and archive seasonal hunts without modifying application source code.
- Built a responsive React and TypeScript frontend integrating dynamic content rendering, interactive progress tracking, image storage, and access-code-protected bonus activities.

HandTune | Python, MediaPipe, OpenCV, PySide6, MIDI | GitHub September 2026
- Engineered a real-time computer vision application using Python, MediaPipe, and OpenCV to translate hand landmark coordinates into continuous audio control signals for instrumental volume and vocal pitch correction.
- Implemented distance-normalized gesture detection, temporal smoothing algorithms, and hand-loss handling to reduce tracking noise and maintain stable, responsive controls across varying camera distances.
- Integrated MIDI Control Change (CC) messaging with MAutoPitch for real-time vocal processing and developed a PySide6 desktop interface featuring webcam visualization, gesture feedback, and interactive playback controls.

Walk2Feed | React native, Expo, TypeScript | Devpost | GitHub November 2025
- Collaborated on a gamified mobile fitness application using React Native, Expo, and TypeScript, integrating step-tracking functionality, personalized activity dashboards, and virtual pet progression mechanics.
- Implemented activity-tracking features that translate mobile sensor data into in-game rewards and integrated **AI-powered conversational and voice interactions** as part of a functional hackathon prototype.

Blowfish Budgeting | React, Node.js, TypeScript | Devpost | GitHub | Live Demo May 2025
- Developed a full-stack personal finance application using React, Node.js, and TypeScript to generate personalized spending plans based on user income and financial goals.
- Implemented financial calculation logic for expense tracking, savings allocation, and budget variance analysis, dynamically updating spending insights through a responsive web interface.

LEADERSHIP & INVOLVEMENT

Legoland Discovery Center Bay Area Milpitas, CA
Guest Experience Host *January 2024 - Present*
- Identified an operational challenge and independently proposed a software solution, collaborating with management to define requirements and secure deployment approval.
- Supported hundreds of daily guests through customer-facing operations and contributed to guest satisfaction **improvements of over 20%**.

Women in Computer Science De Anza Cupertino, CA
Vice President & ICC Representative *June 2025 - June 2026*
- Coordinated technical workshops, professional development events, and student outreach initiatives to support women pursuing computer science.